

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 6 years building Node.js/TypeScript services and scalable architectures for high-performance web and mobile delivery. At MTailor (US e-commerce, reporting to the CTO), led a zero-downtime migration of 20+ production applications to Firebase/GCP, built a real-time two-way data sync on Google Pub/Sub, and shipped revenue features end-to-end (+\$40/order Express Shipping; 3D visualization that lifted conversion). Agency-tested: delivered for demanding US clients at Mind2matter and led a 4-developer team building client systems at CodeBits. Designs secure RESTful APIs, owns work end-to-end, and mentors through documentation and training.

CORE COMPETENCIES

Node.js & TypeScript

Scalable Backend Architecture

Secure RESTful APIs

High-Performance Mobile Delivery

GCP & Firebase

Agile Delivery

Technical Ownership & Mentoring

WORK EXPERIENCE

MTailor (US Remote)

Sep 2022 – Present

Software Engineer — reporting directly to the CTO

- Led the zero-downtime migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP, reporting directly to the CTO
- Implemented **real-time two-way sync** between MongoDB and Firestore using Node.js and Google Pub/Sub for message processing
- Shipped **Express Shipping end-to-end**, generating an additional **\$40 revenue per order**
- Built a **3D visualisation feature** using video overlay and ffmpeg, increasing buyer conversion
- Migrated file storage from Amazon S3 to GCS; moved all services off AWS, saving **\$5,000/month**
- Migrated a Webflow website from a Parse backend to Firebase

MTailor (US Remote)

Mar 2022 – Jul 2022

Contractor (Backend Migration)

- Hired via Upwork onto an unfamiliar stack; built MongoDB→Firestore migration scripts with complex processing logic
- Wrote Firebase SDK documentation (Node.js and Python) other engineers onboarded from; trained the Ops team — converted to full-time and given the company-wide migration lead within months

CodeBits (Uganda)

Jan 2020 – Jul 2021

Led a team of 4 developers building legal tech for NGOs end-to-end

- Owned client relationships and delivery for FIDA Uganda and LASPNET — requirements, architecture, build, ship
- Architected an event-driven microservices backend using **Apache Kafka, Docker, and Kubernetes**
- Built a **USSD service** for legal aid over feature phones, communicating with a Node.js backend via gRPC — meeting rural users on the infrastructure they actually have
- Developed a cross-platform mobile app using **React Native and Expo** (iOS & Android) with Firebase push notifications

Dr Wealth (Singapore Remote)

Aug 2021 – Jan 2022

Software Engineer

- Led optimisation of a PWA delivering real-time stock prices; built jobs querying 2M+ Firestore records from Morningstar APIs
- Extended backend APIs with Firebase Cloud Functions and Express.js; built responsive React/Tailwind UI pages

Mind2matter (US Remote)

Jan 2022 – Jul 2022

Software Engineer (Full-Stack)

- Built backends for DeFi applications using Web3 and Node.js; delivered pixel-accurate React UIs from Figma under tight timelines

PROJECTS

career-ops — Agentic Job-Search Pipeline (open source)

Agentic pipeline built on Claude Code: Playwright ground-truth verification, batch evaluation workers, automated tailored PDF generation. 100+ offers evaluated.

Claude Code · Playwright · Node.js · YAML · Markdown

Healthcare WhatsApp Chatbot (LLM in production)

Integrated OpenAI + Google Vision for prescription OCR via WhatsApp Business API.

Node.js · OpenAI · Google Vision · WhatsApp Business API

Marketplace Recommendation Engine

Collaborative + content-based filtering engine in MongoDB for service-provider matching in a live marketplace.

Node.js · MongoDB

EDUCATION

BSc Computer Engineering — [Makerere University](#), Kampala, Uganda

2018 – 2022

SKILLS

Proficient: Node.js, TypeScript/JavaScript, React, Firebase, GCP, Python, MongoDB, SQL, Git, Linux **Intermediate:** Go, AWS, Rust, React Native

Infrastructure: Kubernetes, Docker, Kafka, gRPC, Pub/Sub **Languages:** English, Kiswahili